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Santa Clara University	ELEN 164 Winter 2026	Shoba Krishnan
Laboratory #7: Studying converter performance		

I. Objectives

- To evaluate the performance of a switching regulator.
- To understand steady state behavior.
- To observe differences in real vs. ideal components

II. Laboratory procedure

Part 1: Working with the board to study converter performance

Repeat the setup you did in section 1 and 2 in Part2 of the previous lab.

Section 1: Setting-up the Buck Configuration
Section 2: Measurement Setup

Section 3: Open Loop Buck Mode Measurements

1. Turn on the power supply output
   a. You will see a red LED light up for a few seconds – that is OK! It is part of the startup sequence. If the LED stays on for more than 5 seconds, turn off the power supply and alert your TA.
   b. If there is an orange LED, that is also OK. That is just a warning to notify you that the voltage on the output exceeds 7V. One of the on-board load resistors cannot exceed 7V, so the LED simply notifies you that the voltage is too high to connect this resistor.

Section 4: Varying Duty Cycle

- Set the switching frequency to 30kHz
- Set the input voltage at 12V
- Fill out the provided chart below by adjusting the potentiometers on the board Note that you can find the average voltage across the load by using the multimeter.

Duty Cycle	Vout (Measured)	Vout (Theoretical)	Vout ripple (Measured)	Vout ripple (theoretical)
42%	4.316V	5.04V	312.5mV	~40mV
50%	5.178V	6V	318.75mV	~41mV
60%	6.22V	7.2V	283.25mV	~39.4mV

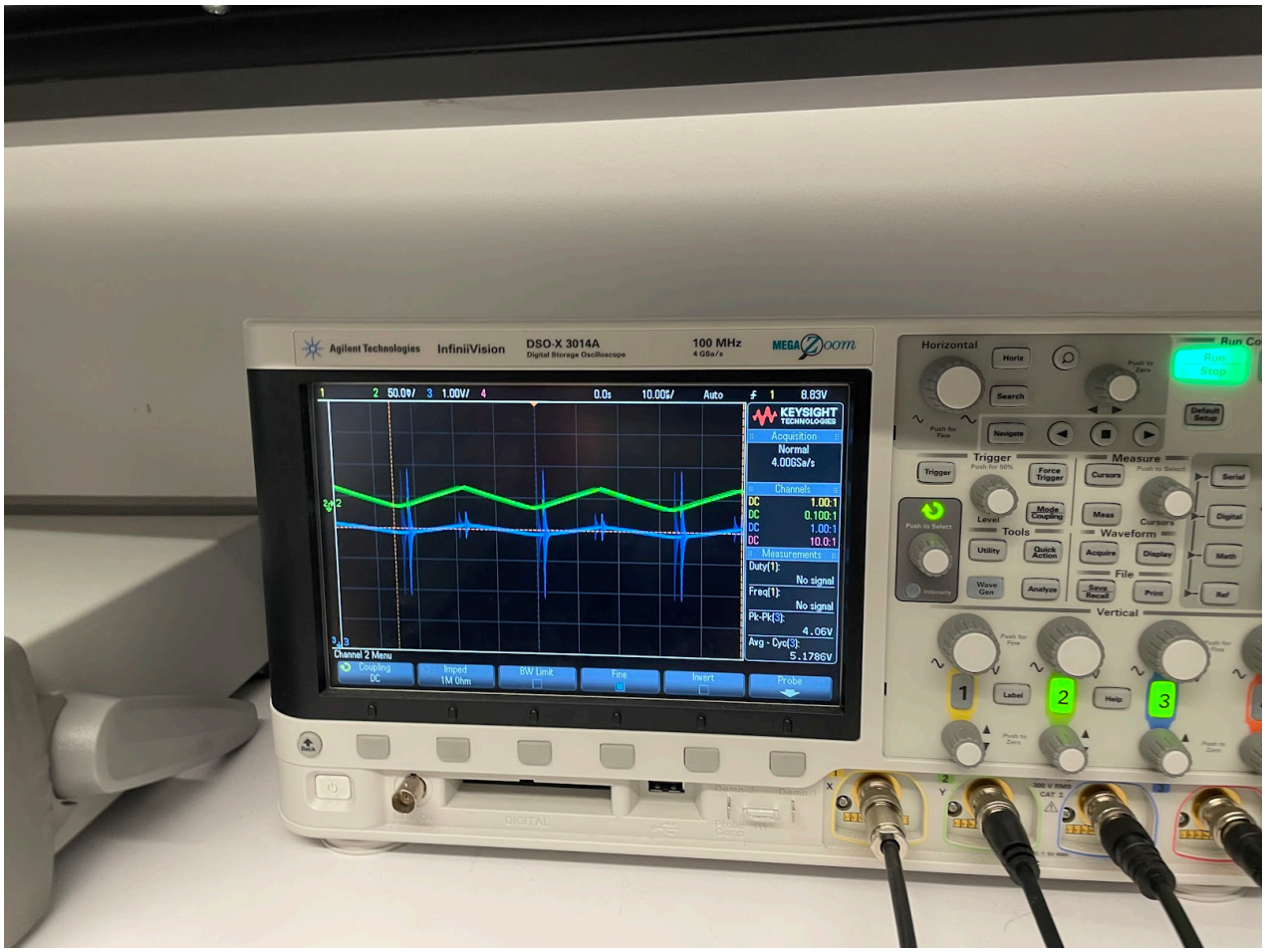


Figure 1: Output current and voltage DC = 0.42

Observations:

- **Do your measurements make sense? Why or why not? Explain your observations clearly.**

Yes they do. Our measured values follow the same general pattern as the theoretical, but vary on a duty cycle by duty cycle basis due to real parasitics.

Section 5: Varying Switching Frequency

- Keep the Duty Cycle at 42%
- Keep the input voltage at 12V
- Fill out the provided chart below by adjusting the potentiometers on the board
    - You can use cursors to measure the ripple.
    - Use the "AC Coupling" option to get rid of the DC Bias on the output.

Switching Frequency	Vout (AVG) (Measured)	Vout Ripple (Measured)	Vout Ripple (Theoretical)
30 KHz	4.316V	312.5mV	~40mV

25 KHz	4.171V	325mV	~57.5mV

Observations:

- **Do your measurements make sense? Why or why not? Explain your observations clearly.**

Yes they do. The measured Vout Ripple on 25kHz is a little low, but still increased when compared to the theoretical.

Section 6: Varying Output Capacitance

- Keep the switching frequency at 30kHz
- Keep the input voltage at 12V
- Fill out the provided chart below by adjusting the output capacitor jumpers on the board
  - You can adjust the capacitance by changing the jumpers on the output capacitors as shown in **the board diagram from Lab 6 or in Figure 1 in the next section.**
    - Note that the board should be fully powered down whenever you adjust the output capacitor jumper settings!
- Take an FFT of the output signal. Observe and record the peak maximum frequency content on the output for each configuration.
  - Where is most of the frequency content and why?
  - Does the FFT change based on the output capacitance?

Output Capacitance	Vout (AVG)	Vout Ripple	Frequency with peak value
No Output Cap	4.2V	900mV	29.92kHz
47 µF	4.316V	312.5mV	29.94kHz
470 µF || 47 µF	4.32V	93.7mV	29.85kHz

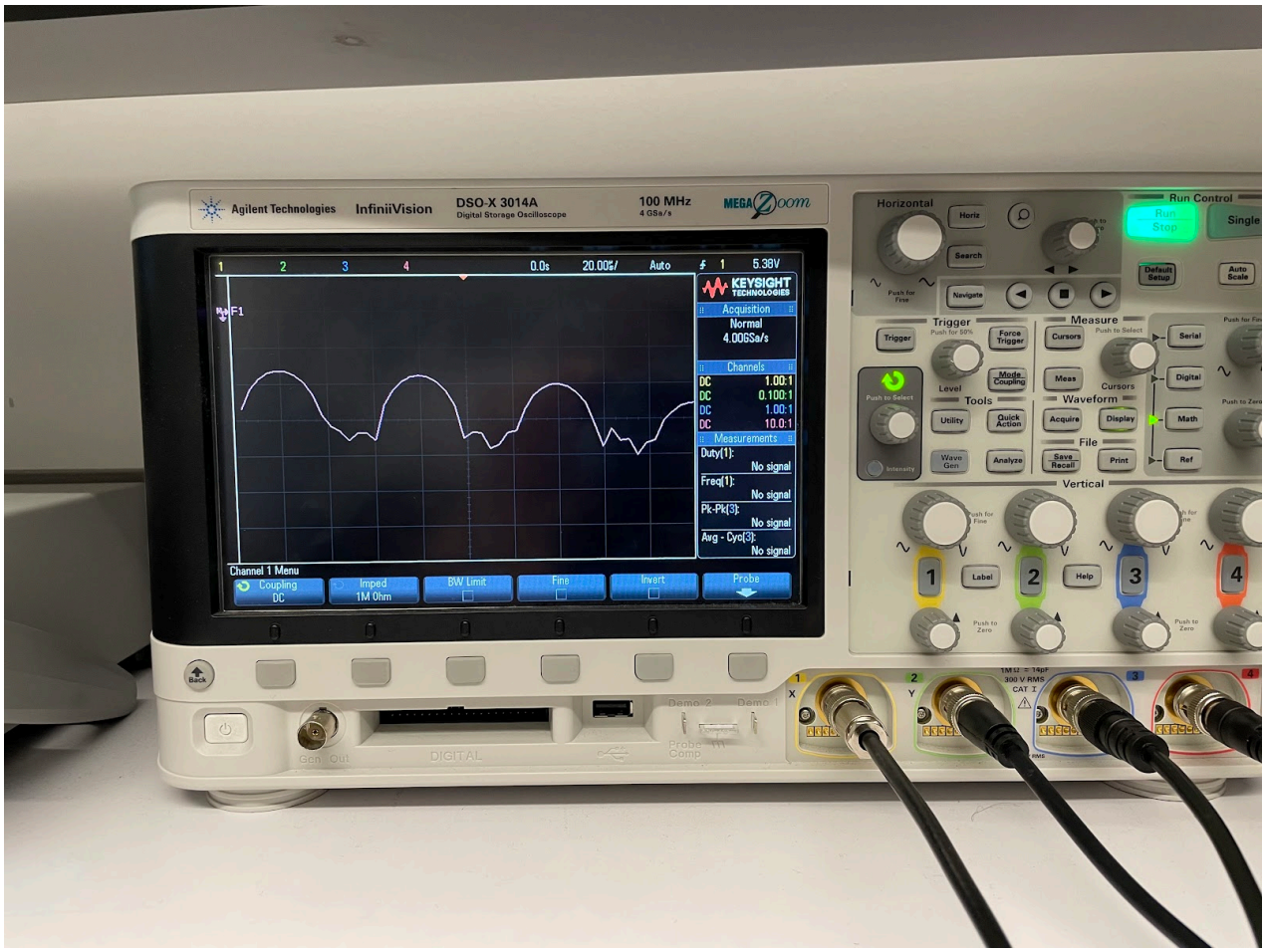


Figure 2: FFT No Output Capacitance

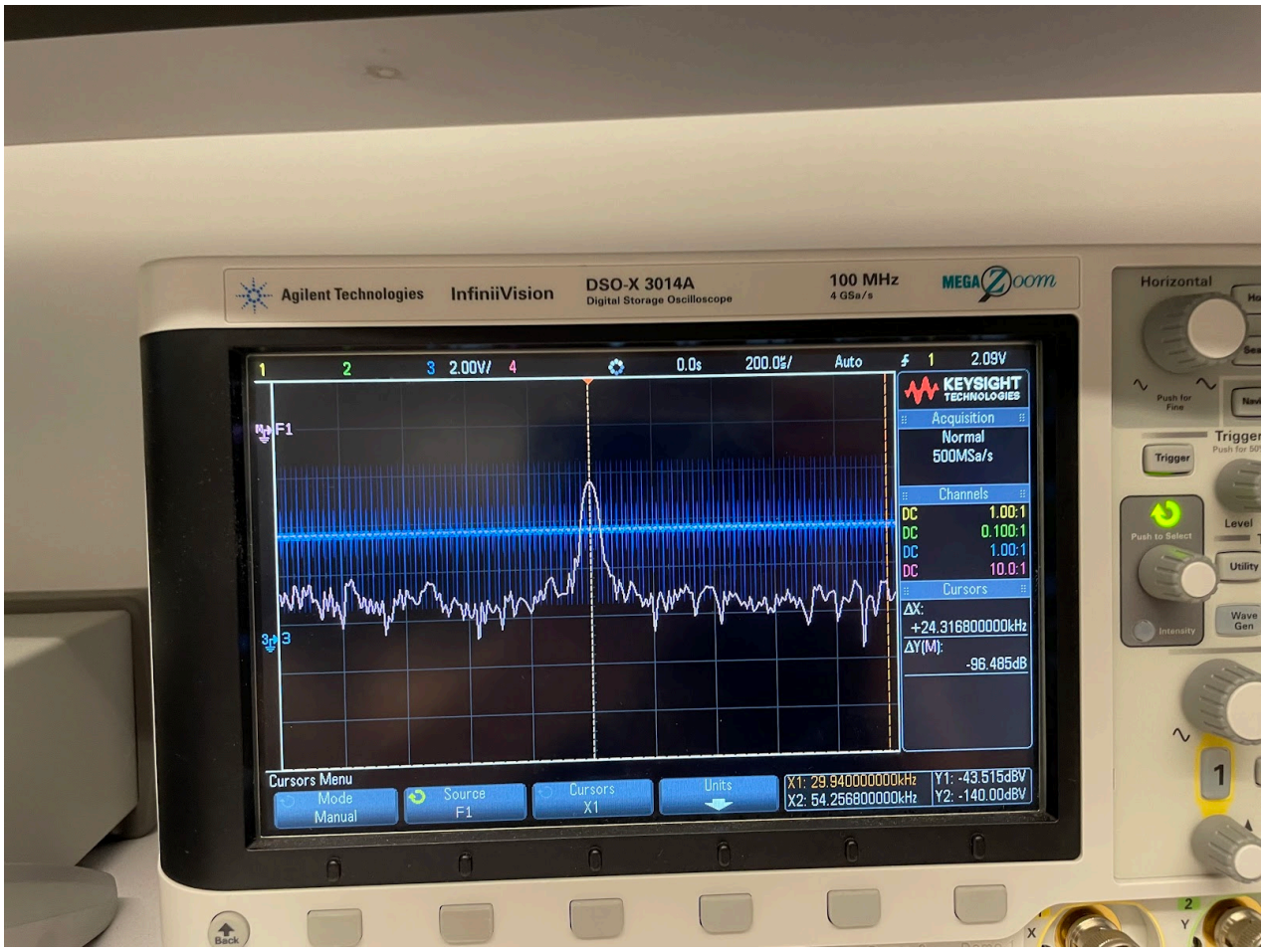


Figure 3: FFT 47uF Capacitance

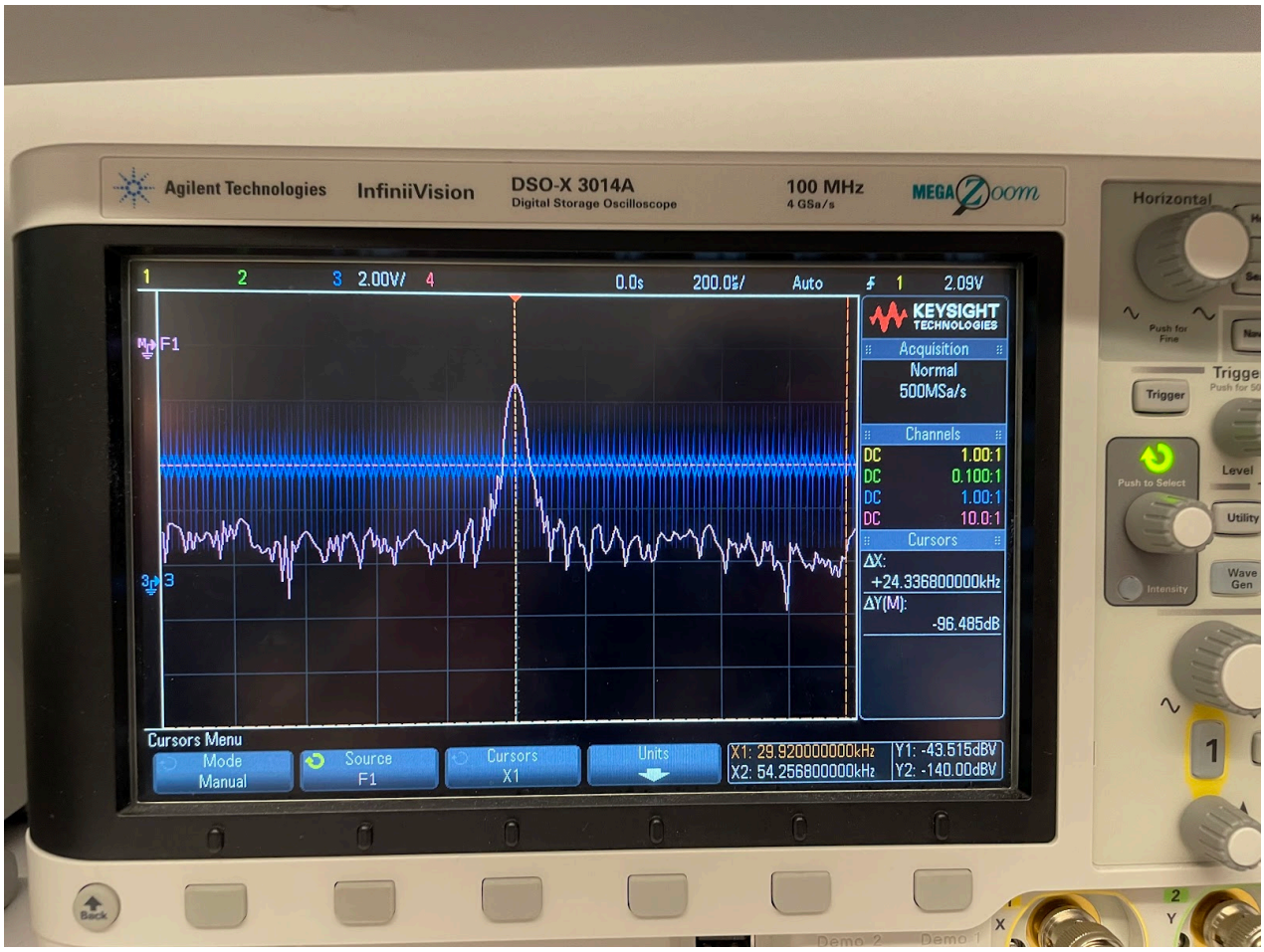


Figure 4: FFT 47uF and 470uF in parallel

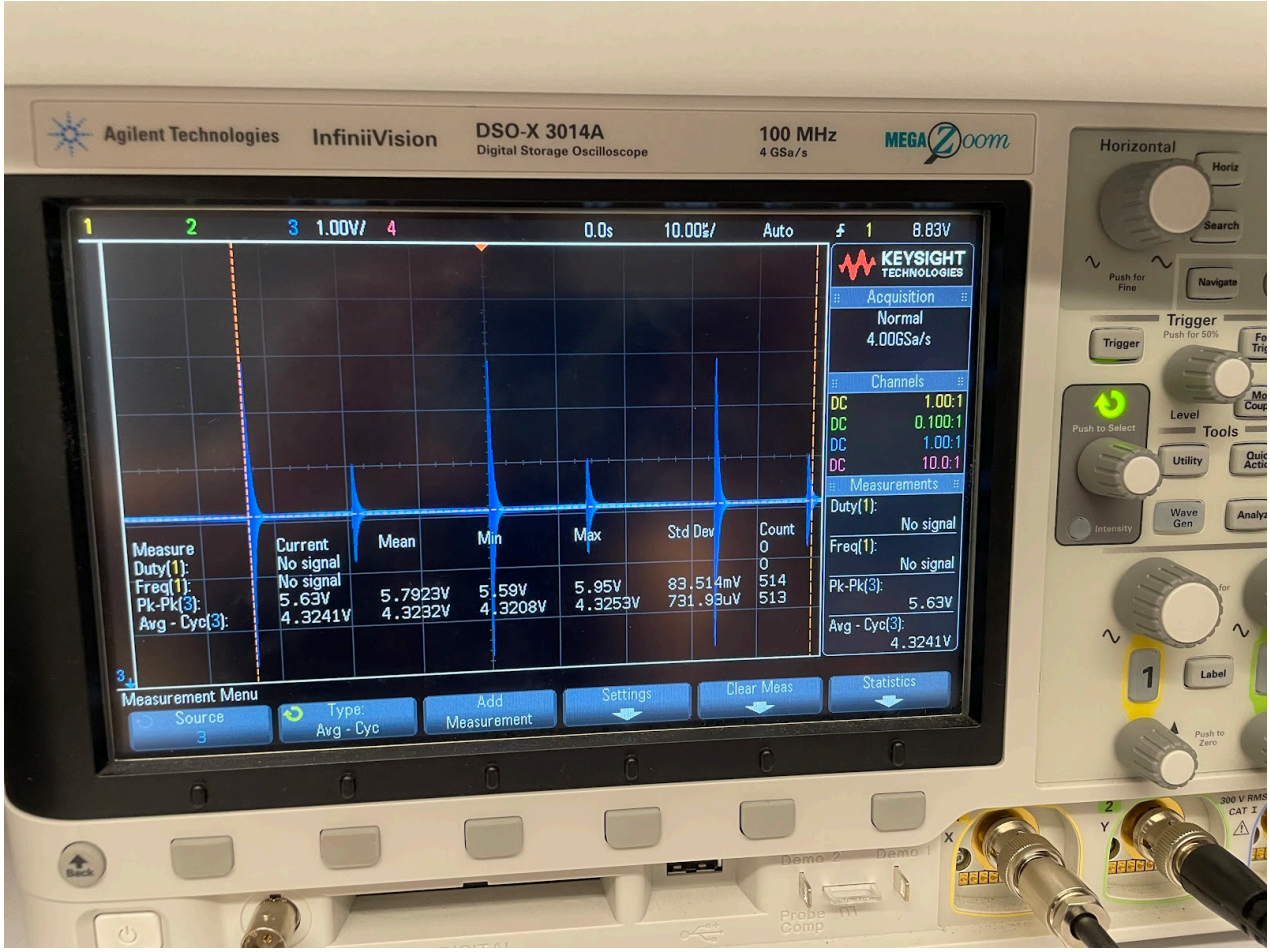


Figure 5: Output voltage of 47uF and 470uF Capacitors in parallel

Observations:

- **Do your measurements make sense? Why or why not? Explain your observations clearly.**

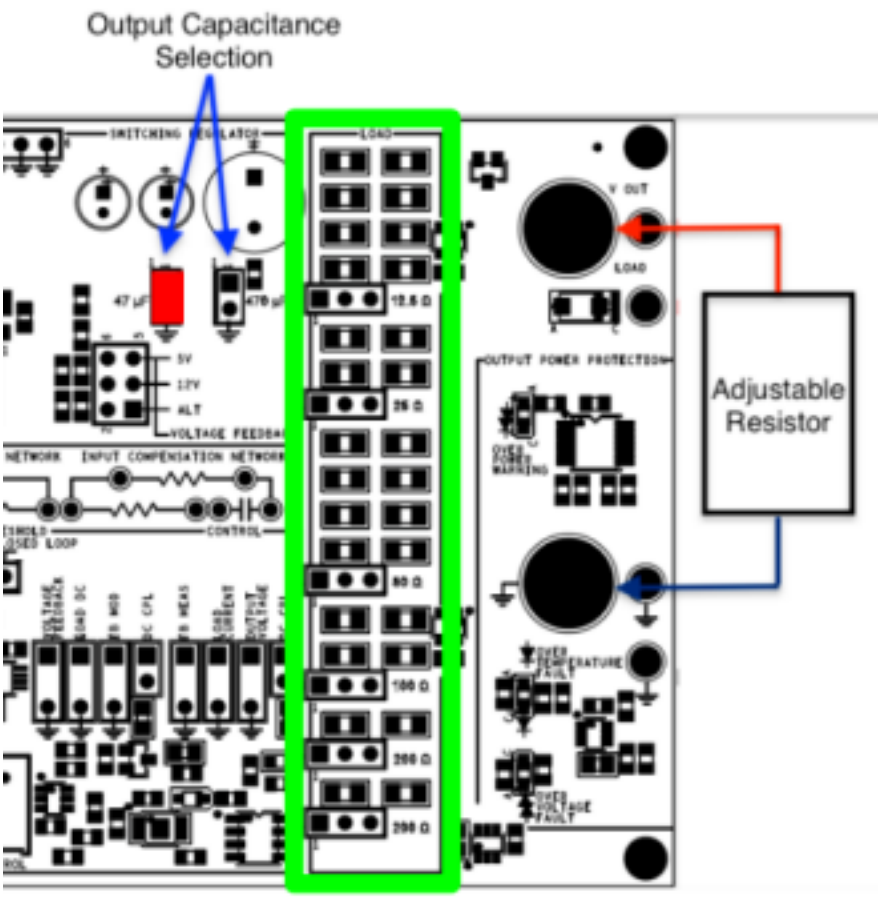
Section 7: Entering DCM Mode

- Keep the switching frequency at 30kHz
- Keep the input voltage at 12V
- Keep the duty cycle at 42%

1. Remove all of the resistor jumpers highlighted on **Figure 1** below.

Note that the board should be fully powered down when you adjust the output resistor settings!

Figure 1: Output Resistance Jumper Selection



2. Attach the large adjustable resistor at your lab station.
   Use the DMM at your station to set your potentiometer to approximately 15 ohms to ensure the converter is in CCM.
3. Slowly turn the potentiometer clockwise and observe the inductor current waveform. Continue to slowly increase the resistance until you see that the converter has entered DCM.

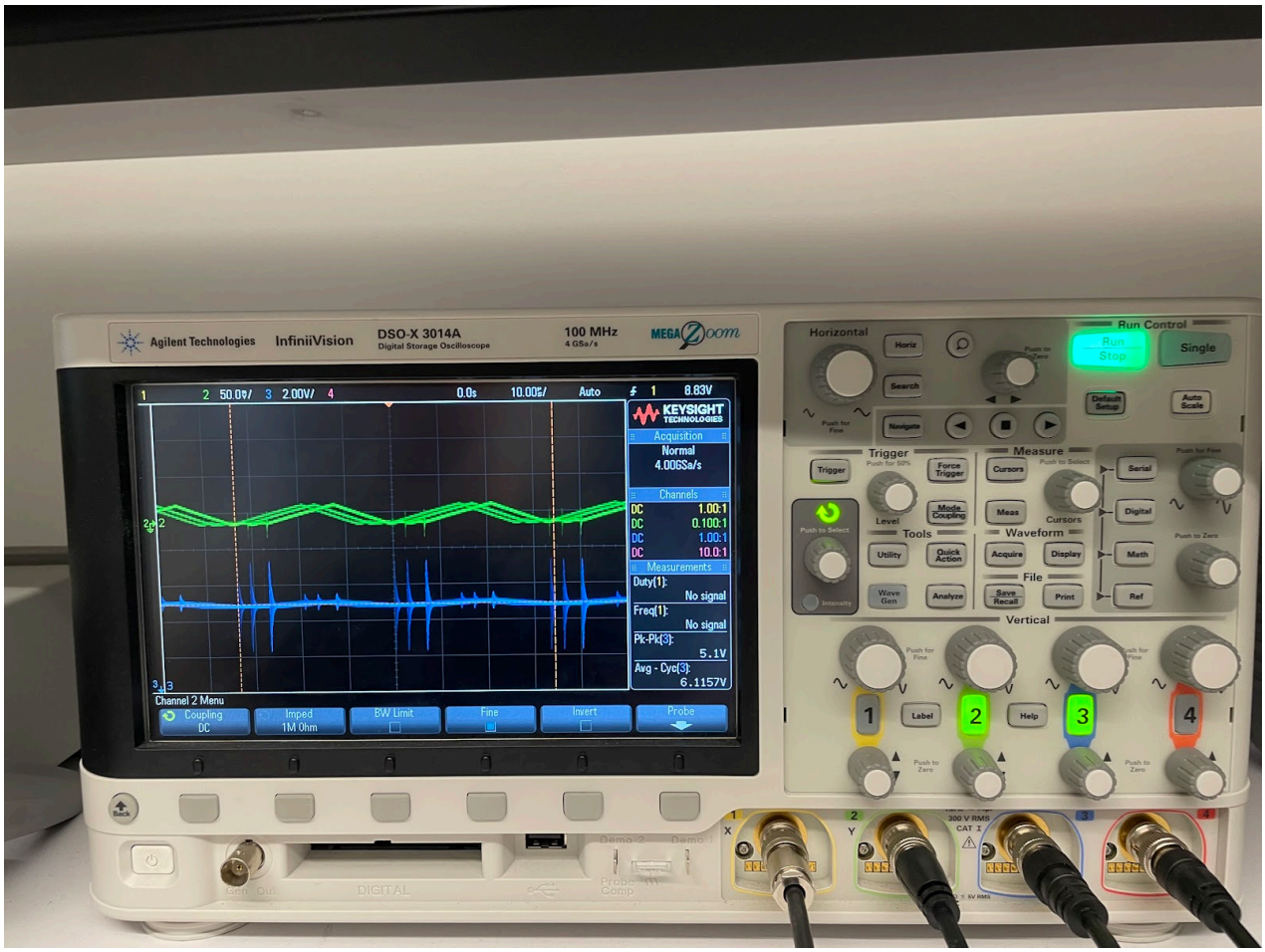


Figure 6: DCM output current

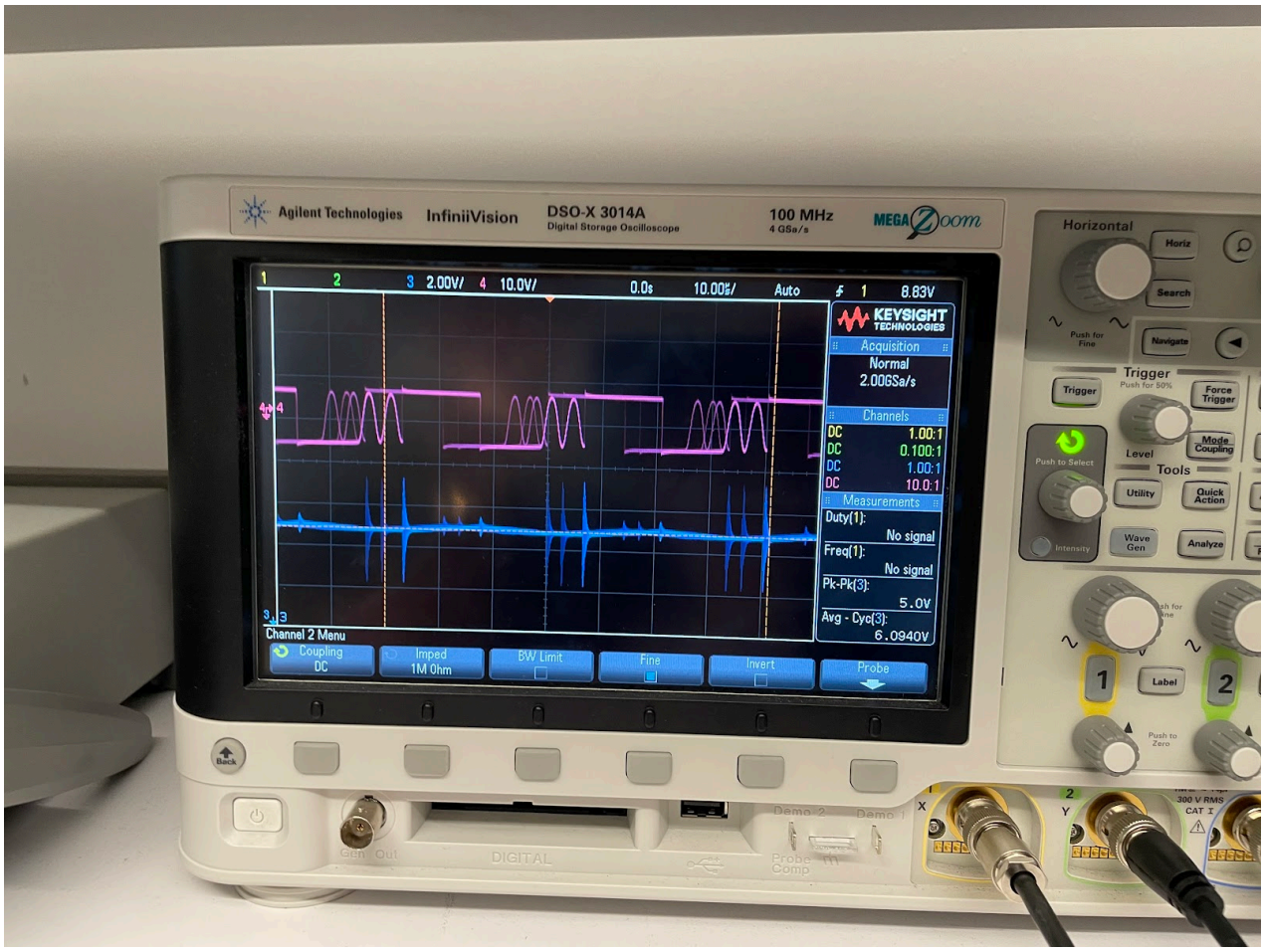


Figure 7: DCM output voltage

4. **Power down the board and unplug the external potentiometer. Use the DMM at your lab station to measure the load resistance that puts this converter into DCM.**

R_DCM = 27.557ohms

Observations:

Does this value agree with your calculations from the prelab from last week?

Yes, within reason. The prelab calculated closer to 22, but that was for ideal components.

5. Reconnect the resistance. Power up the board again and increase it further to ensure the converter is well into DCM.
6. Take snap shots of the inductor current, inductor voltage.
7. Measure the average output voltage.

Vout=6V

Do your physical measurements agree with your simulation results? If not, explain a couple reasons why.

They mostly do, but are a little off due to parasitics.